

Phishing Email Investigation

Sender identity, link behaviour and domain reputation

A personal investigation of a Microsoft account alert

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Lab scope and investigation method

Objective: identify impersonation, inspect message destinations and support a verdict with recorded evidence.

I inspected one public phishing sample as text in **Notepad**, comparing sender identities, recorded authentication results, HTML links and external image references. I then consulted existing domain reports in **VirusTotal**.

Component	Lab configuration
Source sample	rf-peixoto/Phishing Pot, email/sample-10.eml; pinned source linked below.
Message date	8 September 2023; external SMTP handoff recorded at 05:47:04 UTC.
Analysis record	5 October 2026; VirusTotal analysis ages retained as displayed.
Tools and content	Notepad and VirusTotal; HTML message with no attachments.
Inspection method	Static text review; embedded links and the external image were not opened.

Procedure

- 1 Identity and authentication.** Compared From, Reply-To and Return-Path; read Authentication-Results and the receiving gateway's external Received entry.
- 2 Destinations.** Inspected all three href links and the hidden image reference. Distinguished the sending server IP from sign-in details claimed in the body.
- 3 Reputation.** Recorded the existing reports for the three domains and preserved two supplied screenshots. No unique tracking URL was submitted for scanning.
- 4 Assessment.** Combined the observations, documented the unknowns and proposed response actions. Mailbox searches and remediation were not performed.

Scope. This is a static investigation of an archived, anonymised message. No organisational mail-flow, click, reply or account sign-in records were available. The original recipient placeholder is phishing@pot.

Pinned source sample - [rf-peixoto/Phishing Pot](#)

Sender identity and email authentication

Separate the visible sender, envelope sender and reply destination before evaluating authenticity.

SELECTED HEADER VALUES - DEFANGED TRANSCRIPTION

```
Display name:      Microsoft account team
From address:      no-reply@access-accsecurity[.]com
Reply-To:          sotrecognizd@gmail[.]com
Return-Path:       bounce@thcultarfdes[.]co.uk
SMTP MAIL FROM:    thcultarfdes[.]co.uk
Sending server IP: 89.144.44.2
```

Sender comparison

Microsoft documents **account-security-noreply@accountprotection.microsoft.com** as its account alert sender. [1] The sample claims Microsoft branding but uses a different From domain, and its Reply-To leads to a separate Gmail mailbox. Together with the message behaviour, this supports brand impersonation.

The different SMTP MAIL FROM and visible From domains are context, not sufficient proof of phishing by themselves. The source IP is recorded in the receiving gateway's headers; it does not identify a person or prove an account sign-in.

Authentication results recorded in the headers

Check	Recorded result	Interpretation
SPF	none	No SPF policy found for the SMTP sender domain. [2]
DKIM	none	No signature found; not a failed signature check. [2]
DMARC	permmerror	Permanent evaluation error; the exact cause is not shown. [3]

Interpretation. These results are not equivalent to fail. The DMARC header also records action=none. No pass is shown for these three checks, but authentication alone does not establish phishing.

The report preserves the receiver's historical results. DNS authentication was not re-evaluated and the exact cause of DMARC permmerror is unknown.

Links and claimed sign-in activity

Inspect the actual destination. A security warning is not evidence that the stated sign-in occurred.

Visible label	Observed behaviour
phishing@pot	mailto: to the Reply-To mailbox, also in CC; anonymised account label.
Report The User	mailto: to the same mailbox, also in CC; prefilled subject and body.
click here	mailto: to the same mailbox, also in CC; an unsubscribe subject.

All three action links prepare an email draft to the Reply-To mailbox. A `mailto:` link does not automatically send a message. No web sign-in destination or password request was observed in the sample.

Hidden external image

HTML IMAGE REFERENCE - DEFANGED, PATH ABBREVIATED

```
Host:      thebandalisty[.]com
URL prefix: hxxp://thebandalisty[.]com/track/...
Dimensions: width="1px" height="1px"
Style:     visibility:hidden
```

The hidden image is consistent with a tracking pixel if fetched. No image download was observed. Proxies and automated scanners can make a request an unreliable indicator of a human opening the email.

Two IP addresses with different roles

IP address	Role in the evidence
89.144.44.2	External sending server recorded in the receiving gateway's headers.
103.225.77.255	A sign-in IP claimed in the body, alongside Russia/Moscow; not independently verified.

Behaviour conclusion. A reply-based lure with possible tracking. No credential theft or account compromise is established; the claimed sign-in remains unverified.

Domain reputation and evidence

Existing VirusTotal reports recorded on 5 October 2026; relative analysis ages are preserved as displayed.

Domain	Malicious detections	Last analysis display
access-accsecurity[.]com	0/91	2 months ago
thcultarfdes[.]co.uk	0/91	26 days ago
thebandalisty[.]com	9/91	4 days ago

For the hidden image host, seven vendors label **Phishing** and two label **Malicious**. **ESET: Suspicious** is separate from the nine red results. Community score -44 is a separate metric. [4]

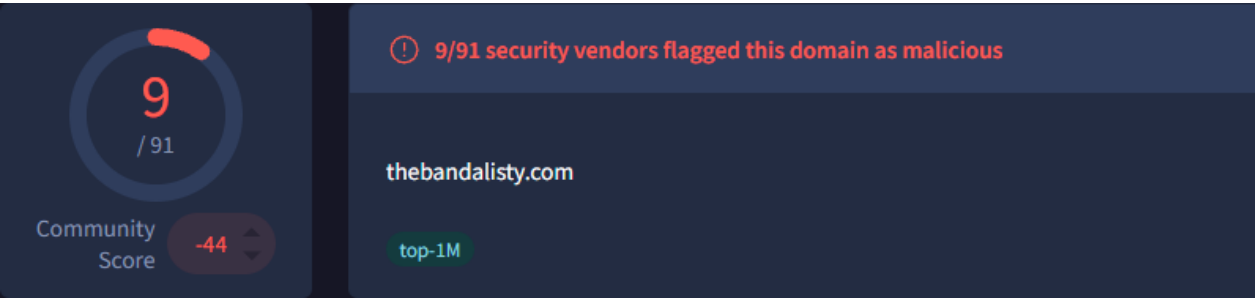


Figure 1. Domain and score excerpt from the supplied VirusTotal screenshot.

BitDefender	⚠ Phishing
Fortinet	⚠ Phishing
Lionic	⚠ Phishing
VIPRE	⚠ Phishing
ESET	ⓘ Suspicious

Figure 2. Four red results and ESET's separate Suspicious classification. The complete screenshot retains all nine red results.

Reputation limit. A zero detection count does not establish safety. The 2026 results do not establish reputation when the email was sent in 2023 or prove successful user interaction.

Full screenshots for the visible sender and hidden image domain accompany the repository. Figures above are rectangular crops with unchanged pixels. The SMTP sender result was recorded from report text; no screenshot was supplied.

Verdict, response and references

An evidence-supported classification, proposed response and a clear boundary around unavailable data.

Verdict: phishing with Microsoft brand impersonation. The message claims to be an account security notification but routes replies and all three action links to an unrelated Gmail mailbox. No credential capture or account compromise was established.

Recommended response

- 1 Preserve and report.** Avoid replying, clicking or loading remote images. Preserve the original email and report it through the organisation's phishing process.
- 2 Contain confirmed messages.** Quarantine or remove matching messages after evidence retention. Apply reviewed, targeted blocks; blocking all Gmail traffic would be too broad.
- 3 Check scope and interaction.** Use mail-flow records to find recipients. Use appropriate reply, click and account activity records to assess interaction and impact. Secure an account if exposure or suspected compromise warrants it.

Uncompleted incident scope. Message Trace shows delivery status, not proof of clicks. [5] URL click telemetry, when available, is a separate source. [6] For this mailto-based sample, reply or sent-message records and user confirmation would also matter. These data were unavailable; no zero-impact or recipient-count claim is made.

Source and integrity

Sample: rf-peixoto/Phishing Pot, pinned revision below. The collection anonymises recipient details as phishing@pot; the source licence at this revision is GNU GPL version 3. Original file: 25,597 bytes. The sample was not received in my own mailbox.

097878eab3043ced70fde2ce0fb3b660fb58717e

SHA-256: 4fbf4c3d80aba156c59004c12c83ff53

dd64c9cf7b7a6029e98fe1da0760783a

References

- [1] Microsoft - unusual sign-in notifications
- [2] Microsoft Learn - email authentication results
- [3] RFC 7489 - DMARC result meanings, Appendix C
- [4] VirusTotal - domain report fields
- [5] Exchange Online - Message Trace
- [6] Microsoft Defender XDR - UrlClickEvents
- Phishing Pot - pinned sample and collection notes
- Phishing Pot - upstream licence at the pinned revision